COVARIANCE DECAY FOR GAUSSIAN QUEUES

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Abstract. We study the large-lag cross-covariance

$$\mathcal{C}(r) = \mathbb{C}\mathrm{ov}\big(Q_{W_1}(0), Q_{W_2}(r)\big)$$

of stationary workloads Q_{W_1} and Q_{W_2} in two Gaussian queues driven by a common input X and drained at possibly different rates $c_1, c_2 > 0$.

When X is a standard Brownian motion, we derive an exact representation in terms of a drifted Brownian hitting time and obtain the sharp asymptotics

$$\mathcal{C}(r) \sim K_{12} r^{-3/2} e^{-c_2^2 r/2} \quad \text{as } r \to \infty,$$

with an explicit positive constant K_{12}.

For Gaussian inputs with stationary increments and regularly varying variance curvature, we prove that the workload covariance inherits the dependence structure of the input process. Specifically, for fractional Brownian motion (fBM) input with Hurst parameter $H \in (0,1) \setminus \{1/2\}$ we get

$$\mathcal{C}(r) \sim H(2H-1)\mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\} r^{2H-2},$$

where τ_{W_i} denote the leftmost time at which $W_i(t) = X(t) - c_i t$ attains its maximum over $[0, \infty)$. A similar result holds for an integrated stationary Gaussian input with positive, regularly varying covariance. For inputs with exponentially decaying variance curvature, we derive explicit exponential covariance asymptotics under a sufficient condition on the later queue's service rate. As an illustrative example, we consider the model with an integrated Ornstein–Uhlenbeck input.

The proofs combine Price's interpolation formula with localisation and tail estimates for the maximising times.

1. Introduction

Consider an infinite-buffer queue driven by $X(t),\ t \in \mathbb{R}$, a Gaussian process with stationary increments and continuous sample paths and drained at a constant rate $c > 0$. Let $W(t) = X(t) - ct,\ t \in \mathbb{R}$ and assume that $W(t) \xrightarrow{t\to\infty} -\infty$ almost surely. Under the interpretation that $X(t) - X(s)$ is the amount of *fluid* that enters the system in time interval $(s, t]$ for $t > s$, the stationary workload process $Q_W(t),\ t \in \mathbb{R}$ admits the representation

$$Q_W(t) := \sup_{u \geq 0} \big(W(t) - W(t-u)\big), \qquad t \in \mathbb{R}; \tag{1}$$

see, e.g., [20]. We refer to [21] or the monograph [16] for functional central limit theorems that provide strong theoretical justification for modelling the input stream X as a Gaussian process.

An important class of models is obtained by taking $W(t) = B_H(t) - ct$, where B_H is a two-sided fractional Brownian motion (fBM) with Hurst parameter $H \in (0,1)$ and $c > 0$. This defines the classical stationary storage model fed by fBM, which is supported by the statistical evidence of the presence of *long-range dependence* and *self-similarity* in input streams [23]; see also [18].

Gaussian queues have attracted substantial interest over the past several decades, particularly in the analysis of stationary buffer content (1). Although the steady-state marginal distribution is now well understood, much less is known about the dependence structure of the workload process. A question raised in [10] asks for the large-time behaviour of

$$\rho(t) = \mathbb{C}\mathrm{orr}(Q_W(t), Q_W(0)), \qquad t \to \infty, \tag{2}$$

and more generally whether the dependence range of the Gaussian input is inherited by the workload process. For fBM input with Hurst parameter $H > 1/2$, an explicit conjecture appears in [17, Conjecture 7.1]; see also

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[10]. It states that

$$t^{2-2H}\,\mathbb{C}\mathrm{ov}(Q_W(0),Q_W(t))\sim Const>0$$

as $t\to\infty$. Related works [9, 11] study workload dependence for fBM and integrated Ornstein–Uhlenbeck input using a different dependence metric, i.e.

$$\mathbb{P}\{Q_W(0)>p,Q_W(t)>q\}$$

under specific relations between t,p,q.

The asymptotic behaviour of (2) depends sharply on the input. The Brownian case $H=1/2$ is Markovian and exceptional, since it decays exponentially to 0 as $t\to\infty$. Specifically, it was shown in [1] that

$$\mathbb{C}\mathrm{ov}(Q_W(0),Q_W(t))\sim\frac{2^{3/2}}{\sqrt{\pi}\,c^5}t^{-3/2}e^{-c^2t/2},\qquad t\to\infty.$$

The above asymptotic relation follows from the representation

$$\mathbb{C}\mathrm{ov}(Q_W(0),Q_W(t))=\int_0^\infty se^{-2cs}\mathbb{P}\{T_s^{(c)}>t\}\,ds,\quad t\geq 0, \tag{3}$$

where $T_s^{(c)}$ is the hitting time of zero by a Brownian motion with drift $-c$ started from s. For fBM input with $H\neq 1/2$, no Markovian representation such as (3) is available and thus a different approach is needed.

In this paper, we analyse the dependence structure of a more general model comprising two queues in parallel. Specifically, we consider a two-node system where both queues are driven by a common input process X and drained at rates c_1 and c_2, respectively. Then, our objective is to find the exact asymptotics of

$$\mathbb{C}\mathrm{ov}(Q_{W_1}(0),Q_{W_2}(t)) \tag{4}$$

as $t\to\infty$, where $W_i(t)=X(t)-c_it$. The joint steady-state distribution of such a model, in the Brownian setting, was considered in e.g., [15].

As we show below, the asymptotics of (4), as $t\to\infty$, is closely related to the asymptotics of the cross-covariance of two distant increments of the input process X. Then, the *Price interpolation* technique, developed in Section 3, allows us to transfer this leading kernel to the two maximising locations, yielding the required asymptotics of the buffer content covariance. As a result, we establish the conjectured covariance asymptotics for the fBM queue when $H>1/2$, and show that for $H<1/2$ the covariance has the same polynomial order with a negative leading constant. A similar phenomenon occurs for the integrated stationary Gaussian inputs

$$W_i(t)=\int_0^t Z(s)\,ds-c_it,$$

where Z is a centred stationary Gaussian process with positive covariance function $r_Z(r):=\mathbb{C}\mathrm{ov}(Z(0),Z(r))$. If $r_Z(r)\sim r^{-\alpha}L(r)$ as $r\to\infty$, where $\alpha\geq 0$ and L is slowly varying,and eventually positive, then the workload covariance has the same regularly varying decay rate $r^{-\alpha}L(r)$, under the assumptions stated in Section 2.2. We also consider inputs with exponentially decaying variance curvature, including integrated Ornstein–Uhlenbeck input; see, e.g., [14] for a justification of this model. Their treatment requires exponential-moment estimates for the maximising time of the queue observed at the later time. This yields exponential covariance decay in (4) under a sufficient condition on the queues service rates.

Organisation of the paper. The main results are stated in Section 2; the auxiliary Price interpolation tools are collected in Section 3; the proofs of the results presented in Section 2 are given in Section 4.

2. Main Results

Let

$$W_i(t)=X(t)-c_it,\qquad c_i>0,\qquad i=1,2,$$

where X is a centred Gaussian process with stationary increments, continuous sample paths, $X(0)=0$ almost surely and variance function

$$V_X(t):=\mathbb{V}\mathrm{ar}(X(t)),\qquad t\geq 0.$$

For a continuous process U on $[0,\infty)$ we write

$$M_U(T) := \sup_{0\le s\le T} U(s), \qquad \tau_U(T) := \min \arg\max_{0\le s\le T} U(s).$$

Thus $\tau_U(T)$ denotes the leftmost maximiser. We write M_U and τ_U for the corresponding maximum and leftmost maximiser over $[0,\infty)$ whenever the maximum is finite and attained. Uniqueness of the maximisers for the inputs considered below is established in Section 3.
For $Q_{W_1}(t),\ t\in\mathbb{R}$, $Q_{W_2}(t),\ t\in\mathbb{R}$ defined in (1), we analyse

$$\mathbb{C}\mathrm{ov}(Q_{W_1}(0), Q_{W_2}(r)),$$

as $r\to\infty$.
The crucial observation is that the asymptotic behaviour of the workloads covariance is governed by the covariance between two distant increments of the input

$$R_r(u,v) := \mathbb{C}\mathrm{ov}(X(0)-X(-u), X(r)-X(r-v));$$

see Theorem 3.8 which makes this observation precise. More specifically, if

$$\sup_{0\le u,v\le T}\left|\frac{R_r(u,v)}{b(r)} - g_1(u)g_2(v)\right| \xrightarrow{r\to\infty} 0, \qquad T>0 \tag{5}$$

for $b(r)>0$, $b(r)\xrightarrow{r\to\infty}0$, and continuous functions $g_1,g_2:[0,\infty)\to\mathbb{R}$, then under some mild assumptions stated in Theorem 3.8,

$$\frac{\mathbb{C}\mathrm{ov}(Q_{W_1}(0), Q_{W_2}(r))}{b(r)} \xrightarrow{r\to\infty} \mathbb{E}\{g_1(\tau_{W_1})\}\mathbb{E}\{g_2(\tau_{W_2})\}.$$

We verify these assumptions for two important classes of processes with eventually twice continuously differentiable variance functions. Specifically, under the assumption that (5) holds, we consider processes with:

- ◇ *Regularly varying variance curvature:* V_X'' is regularly varying at infinity with index $-\alpha$, $\alpha\ge 0$, allowing either eventual sign. When $\alpha=0$, we additionally assume that $V_X''(t)\log t\xrightarrow{t\to\infty}0$.
- ◇ *Exponentially decaying variance curvature:*

$$V_X''(t)\sim ae^{-\lambda t}, \qquad t\to\infty, \qquad a,\lambda>0.$$

The above families of Gaussian processes cover important classes of input processes considered in the literature, including: fractional Brownian motions, and integrated stationary Gaussian processes; we refer to the monograph [16] for a theoretic background supproting the use of these processes in modelling of Gaussian queues.
Brownian input requires a separate argument, since R_r vanishes on every fixed compact set for all sufficiently large r, and the leading contribution instead comes from rare, long maximising times. We analyze this case in Section 2.1.

2.1. **Brownian case.** Let $B(t),\ t\in\mathbb{R}$ be a standard two-sided Brownian motion with $B(0)=0$ and take $X=B$. For $a\ge 0$ and $c>0$ we set

$$T_a^{(c)} := \inf\{u\ge 0 : a+B(u)-cu=0\}$$

and

$$U_i := Q_{W_i}(0), \qquad m_i := \mathbb{E}\{U_i\} = \frac{1}{2c_i}, \qquad i=1,2.$$

Theorem 2.1. *For every $r\ge 0$*

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0), Q_{W_2}(r)\big) = \int_0^\infty \mathbb{E}\big\{(U_1-m_1)\mathbf{1}_{\{U_2>a\}}\big\}\,\mathbb{P}\{T_a^{(c_2)}>r\}\,da. \tag{6}$$

Moreover, as $r\to\infty$, we have

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0), Q_{W_2}(r)\big) \sim K_{12}r^{-3/2}e^{-c_2^2 r/2}, \tag{7}$$

where

$$0<K_{12}=\sqrt{\frac{2}{\pi}}\begin{cases}\dfrac{2(2c_2-c_1)}{c_2^6}, & 0<c_1\le c_2,\\ \dfrac{2}{c_2^5}\left(1-\dfrac{c_1-c_2}{\sqrt{(c_1-c_2)^2+c_2^2}}\right), & c_1>c_2.\end{cases}$$

Remark 2.2. At a fixed time, the vector $(Q_{W_1}(0),Q_{W_2}(0))$ has the same law as

$$\left(\sup_{u\ge 0}(B(u)-c_1u),\sup_{u\ge 0}(B(u)-c_2u)\right).$$

Thus it is a two-point instance of the maximum process describing the stationary law of coupled reflected Brownian motions studied in [12]. In particular, [12, Corollary 3.5, equation (3.6)] yields

$$\mathbb{C}\mathrm{ov}(Q_{W_1}(0),Q_{W_2}(0))=\frac{1}{4c_+^2}\left(2-\frac{c_-}{c_+}\right),\qquad c_+=\max(c_1,c_2),\quad c_-=\min(c_1,c_2).$$

Note that the exponential rate in (7) is governed only by the drift c_2 of the queue observed at the later time. Interchanging the roles of c_1 and c_2 in (7) shows that $\mathbb{C}\mathrm{ov}(Q_{W_2}(0),Q_{W_1}(r))$ decays at the rate $r^{-3/2}e^{-c_1^2r/2}$.

2.2. **Regularly varying variance curvature.** We next consider Gaussian inputs with stationary increments whose variance curvature is regularly varying. For real-valued functions, we use the convention that regular variation means a constant nonzero eventual sign and regular variation of the absolute value.

Theorem 2.3. *Suppose that $V_X(t)>0$ for every $t>0$, that V_X is twice continuously differentiable on (t_0,∞) for some $t_0\ge 0$, and that V_X'' is regularly varying at infinity with index $-\alpha$, $\alpha\ge 0$. When $\alpha=0$, assume in addition that $V_X''(t)\log t\xrightarrow{t\to\infty}0$. Then, for every $c_1,c_2>0$,*

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big)\sim\frac{1}{2}\mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\}V_X''(r),\qquad r\to\infty. \tag{8}$$

The following two examples specialise Theorem 2.3 to fBM input and integrated stationary Gaussian input with regularly varying covariance.

Example 2.4. Suppose that $X=B_H$ with $H\in(0,1)\setminus\{1/2\}$, i.e.

$$\mathbb{E}\{B_H(t)\}=0,\qquad \mathbb{C}\mathrm{ov}(B_H(s),B_H(t))=\frac{1}{2}\big(|s|^{2H}+|t|^{2H}-|t-s|^{2H}\big).$$

Then, $V_X(t)=t^{2H}>0$ and

$$V_X''(t)=2H(2H-1)t^{2H-2},\qquad t>0.$$

Thus, Theorem 2.3 applies with $\alpha=2-2H$, leading to

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big)\sim H(2H-1)\mathbb{E}\{\tau_{W_1}\}\,\mathbb{E}\{\tau_{W_2}\}\,r^{2H-2},$$

as $r\to\infty$.
Clearly, this asymptotics also identifies the dependence range of the stationary workload. For every $H\neq 1/2$ we have

$$\mathbb{C}\mathrm{orr}(Q_{W_1}(0),Q_{W_2}(r))\sim\frac{H(2H-1)\mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\}}{\sqrt{\mathbb{V}\mathrm{ar}(M_{W_1})\,\mathbb{V}\mathrm{ar}(M_{W_2})}}r^{-2(1-H)},\qquad r\to\infty.$$

Thus, for $H>1/2$ the workload has positive non-integrable correlation. If $H<1/2$ the correlation is eventually negative and integrable and the workload is short-range dependent in the usual absolute-integrability sense.

Example 2.5. We now consider integrated stationary Gaussian input. Let $Z(t)$, $t\in\mathbb{R}$ be a centred stationary Gaussian process with continuous sample paths and covariance function $r_Z(r):=\mathbb{C}\mathrm{ov}(Z(0),Z(r))$. Suppose that

$$X(t):=\int_0^t Z(s)\,ds,\qquad t\in\mathbb{R},$$

with the usual convention $\int_0^t := -\int_t^0$ for $t<0$. Thus, X has stationary increments, continuous sample paths and $X(0)=0$. Assume that $r_Z(t)>0$ for every $t\ge 0$. Suppose that, for some $\alpha\ge 0$ and a slowly varying function L,

$$r_Z(t)\sim L(t)t^{-\alpha},\qquad t\to\infty, \tag{9}$$

and that $L(t)\log t \xrightarrow{t\to\infty} 0$ when $\alpha=0$.
Then,

$$V_X(t)=2\int_0^t (t-s)r_Z(s)\,ds>0,\qquad V_X''(t)=2r_Z(t),\qquad t>0.$$

Thus, by Theorem 2.3,

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big)\sim \mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\}L(r)r^{-\alpha},\quad r\to\infty.$$

2.3. **Exponentially decaying variance curvature.** We now consider inputs whose variance curvature decays exponentially.

Theorem 2.6. *Suppose that $V_X(t)>0$ for every $t>0$, that V_X is twice continuously differentiable on (t_0,∞) for some $t_0\ge 0$, and that, for some $a,\lambda>0$,*

$$V_X''(t)\sim ae^{-\lambda t},\qquad t\to\infty. \tag{10}$$

If

$$c_2^2>2\lambda D_X, \tag{11}$$

where $D_X:=\lim_{t\to\infty}\frac{V_X(t)}{t}$, then for every $c_1>0$,

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big)\sim \frac{a}{2\lambda^2}\mathbb{E}\{1-e^{-\lambda\tau_{W_1}}\}\mathbb{E}\{e^{\lambda\tau_{W_2}}-1\}e^{-\lambda r},\qquad r\to\infty. \tag{12}$$

Remark 2.7. (1) As shown in the proof of Theorem 2.6, both factors $\mathbb{E}\{1-e^{-\lambda\tau_{W_1}}\},\mathbb{E}\{e^{\lambda\tau_{W_2}}-1\}$ are finite and strictly positive. Constnt $D_X\in[0,\infty)$ is well defined, and when $D_X=0$, condition (11) holds for every $c_2>0$. (2) The sufficient condition (11) has a natural large-deviation interpretation. If $D_X>0$, then, for $c>0$,

$$\lim_{t\to\infty}\frac{1}{t}\log\mathbb{P}\{X(t)>ct\}=-\frac{c^2}{2D_X}.$$

The limiting Price kernel contains $e^{\lambda\tau_W}-1$, and (31) ensures

$$\mathbb{E}\{e^{q\tau_W}\}<\infty,\qquad 0<q<\frac{c^2}{2D_X}.$$

Thus $q=\lambda$ is admissible when $c^2>2\lambda D_X$. If $D_X=0$, all positive exponential moments of τ_W are finite. The coefficient a in (10) does not determine D_X: adding an independent Brownian component increases D_X without changing $V_X''(t)$ for $t>0$.

Example 2.8. Let $Z(t),\ t\in\mathbb{R}$ be a centred stationary Ornstein–Uhlenbeck process with covariance

$$r_Z(t)=\sigma^2e^{-\lambda|t|},\qquad \sigma,\lambda>0,$$

and put $X(t)=\int_0^t Z(s)\,ds$, with the usual sign convention for $t<0$. Then

$$V_X(t)=\frac{2\sigma^2}{\lambda}t-\frac{2\sigma^2}{\lambda^2}(1-e^{-\lambda t}),\qquad t\ge 0,$$

so $V_X(t)>0$ for $t>0$,

$$V_X''(t)=2\sigma^2e^{-\lambda t},\qquad a=2\sigma^2,\qquad D_X=\frac{2\sigma^2}{\lambda}.$$

Consequently, Theorem 2.6 applies whenever $c_2>2\sigma$, and yields

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big)\sim \frac{\sigma^2}{\lambda^2}\mathbb{E}\{1-e^{-\lambda\tau_{W_1}}\}\mathbb{E}\{e^{\lambda\tau_{W_2}}-1\}e^{-\lambda r},\qquad r\to\infty. \tag{13}$$

Theorem 2.6 allows us to consider nonhomogeneous models. For example, let B be a standard two-sided Brownian motion independent of Z, and set

$$X(t) = \beta B(t) + \int_0^t Z(s)\, ds, \qquad \beta \geq 0. \tag{14}$$

Then

$$V_X(t) = \beta^2 t + \frac{2\sigma^2}{\lambda} t - \frac{2\sigma^2}{\lambda^2}(1 - e^{-\lambda t}), \qquad t \geq 0,$$

so $a = 2\sigma^2$ and $D_X = \beta^2 + 2\sigma^2/\lambda$. The same covariance asymptotics as (13) therefore holds whenever

$$c_2^2 > 2\lambda\beta^2 + 4\sigma^2,$$

with τ_{W_i} now denoting the maximising times for input (14) respectivelly. Thus the Brownian component changes only the asymptotic constants and the sufficient service-rate condition while leaving the exponentially decaying speed of convergence to 0 unchanged.

3. Auxiliary Results

3.1. **Finite-dimensional and compact Price interpolation.** The finite-dimensional identity presented in the next lemma is the two-block covariance form of Price's theorem; see [19, 22]. Related interpolation representations of Gaussian-maxima variances in terms of covariance kernels appear in [8, Lemmas 3.1 and 3.4]. We establish the compact and half-line extensions needed for the analysis of the asymptotics of (4).

Lemma 3.1. *Let $(\xi, \eta) = ((\xi_i)_{i\in I}, (\eta_j)_{j\in J})$ be a centred Gaussian vector indexed by finite non-empty sets I, J. Put*

$$R_{ij} := \mathbb{C}\mathrm{ov}(\xi_i, \eta_j).$$

Let ξ' and η' be independent centred Gaussian vectors with the same marginal laws as ξ and η, independent of (ξ, η), and set

$$\xi^{(\theta)} := \sqrt{\theta}\,\xi + \sqrt{1-\theta}\,\xi', \qquad \eta^{(\theta)} := \sqrt{\theta}\,\eta + \sqrt{1-\theta}\,\eta'.$$

If $F \in C^1(\mathbb{R}^I)$ and $G \in C^1(\mathbb{R}^J)$ have bounded first partial derivatives, then

$$\mathbb{C}\mathrm{ov}(F(\xi), G(\eta)) = \int_0^1 \sum_{i\in I}\sum_{j\in J} R_{ij}\mathbb{E}\{\partial_i F(\xi^{(\theta)})\partial_j G(\eta^{(\theta)})\}\, d\theta. \tag{15}$$

Proof. First suppose that the joint covariance matrix of (ξ, η) is positive definite and that $F, G \in C^2$ have bounded first and second derivatives. The standard Gaussian interpolation form of Price's theorem, applied to

$$\Phi(\theta) := \mathbb{E}\{F(\xi^{(\theta)})G(\eta^{(\theta)})\}, \qquad \theta \in [0,1],$$

yields, for $\theta \in (0,1)$,

$$\Phi'(\theta) = \sum_{i\in I}\sum_{j\in J} R_{ij}\,\mathbb{E}\Big\{\partial_i F(\xi^{(\theta)})\partial_j G(\eta^{(\theta)})\Big\}. \tag{16}$$

Indeed, the marginal covariance blocks are independent of θ, while the derivative of the cross-covariance block is (R_{ij}). Since $\Phi(1) - \Phi(0) = \mathbb{C}\mathrm{ov}(F(\xi), G(\eta))$, integration of (16) over $[0,1]$ yields the formula (15)..
At fixed positive-definite covariance, the same identity holds for C^1 functions with bounded first derivatives by mollification. The mollified gradients remain uniformly bounded and converge locally uniformly, while the mollified functions have uniform linear growth, so dominated convergence applies on both sides of (15).
It remains to remove the non-degeneracy assumption. Let U, V, U', V' be mutually independent standard Gaussian vectors of the appropriate dimensions, independent of all previously defined vectors, and put

$$\xi_\varepsilon := \xi + \sqrt{\varepsilon}\,U, \qquad \eta_\varepsilon := \eta + \sqrt{\varepsilon}\,V,$$

with independent marginal copies

$$\xi'_\varepsilon := \xi' + \sqrt{\varepsilon}\,U', \qquad \eta'_\varepsilon := \eta' + \sqrt{\varepsilon}\,V'.$$

The covariance matrix of $(\xi_\varepsilon, \eta_\varepsilon)$ is the covariance matrix of (ξ, η) plus $\varepsilon\,\mathrm{Id}$, and is therefore positive definite, whereas its cross-covariance block is still (R_{ij}). Hence the asserted identity holds with $\xi_\varepsilon, \eta_\varepsilon$ in place of ξ, η.

We have

$$\xi_\varepsilon^{(\theta)} \xrightarrow{\varepsilon\downarrow 0} \xi^{(\theta)}, \qquad \eta_\varepsilon^{(\theta)} \xrightarrow{\varepsilon\downarrow 0} \eta^{(\theta)}$$

almost surely for every θ. Since bounded first derivatives make F and G Lipschitz, $F(\xi_\varepsilon^{(\theta)}) \xrightarrow[\varepsilon\downarrow 0]{L^2} F(\xi^{(\theta)})$ and $G(\eta_\varepsilon^{(\theta)}) \xrightarrow[\varepsilon\downarrow 0]{L^2} G(\eta^{(\theta)})$, then the covariance converges. Continuity and boundedness of the first derivatives ensure convergence of the integrand on the right, dominated uniformly on $[0,1]\times\Omega$. Letting $\varepsilon \downarrow 0$ proves the result for arbitrary, possibly singular, Gaussian covariance matrices.
This completes the proof. □

Hereafter E and F are non-empty compact metric spaces, and $C(E)$ denotes the space of continuous real-valued functions on E, equipped with the uniform norm; $C(F)$ is defined analogously. Write δ_u for the unit point mass at u. For $h \in C(E)$, $\gamma > 0$ and a non-empty finite set $D_m \subset E$, we introduce the *softmax* probability measure

$$\pi_{m,\gamma}^h := \sum_{u\in D_m} \frac{e^{\gamma h(u)}}{\sum_{z\in D_m} e^{\gamma h(z)}} \delta_u.$$

Lemma 3.2. *Given $h \in C(E)$ set $A := \arg\max_{u\in E} h(u)$ and let $D_m \subset E$ be nonempty finite sets such that*

$$\Delta_m := \sup_{x\in E} d(x, D_m) \xrightarrow{m\to\infty} 0.$$

For every $\varphi \in C(E)$ that is constant on A, with $\varphi|_A = c$, we have

$$\lim_{m\to\infty} \lim_{\gamma\to\infty} \int_E \varphi(u)\, \pi_{m,\gamma}^h(du) = c.$$

Proof. For fixed m, let $A_m := \arg\max_{u\in D_m} h(u)$. Since D_m is finite and nonempty, A_m is finite and nonempty. As $\gamma \to \infty$, the softmax weights concentrate uniformly on A_m. Hence

$$\lim_{\gamma\to\infty} \int_E \varphi(u)\, \pi_{m,\gamma}^h(du) = \frac{1}{|A_m|} \sum_{u\in A_m} \varphi(u).$$

We shall prove that

$$\sup_{u\in A_m} d(u, A) \xrightarrow{m\to\infty} 0. \tag{17}$$

Let $\varepsilon > 0$ be fixed. If $\{u \in E : d(u, A) \geq \varepsilon\}$ is nonempty, the compactness of E implies that there exists $\delta_\varepsilon > 0$ such that

$$h(u) \leq \max_{u\in E} h(u) - \delta_\varepsilon, \qquad d(u, A) \geq \varepsilon.$$

Let $\tau \in A$. Since $\Delta_m \xrightarrow{m\to\infty} 0$, there are points $z_m \in D_m$ with $z_m \xrightarrow{m\to\infty} \tau$, and hence

$$\max_{z\in D_m} h(z) \geq h(z_m) \xrightarrow{m\to\infty} \max_{u\in E} h(u).$$

For all sufficiently large m we therefore have

$$\max_{z\in D_m} h(z) > \max_{u\in E} h(u) - \delta_\varepsilon.$$

No maximiser over D_m can then have distance at least ε from A. If $\{u \in E : d(u, A) \geq \varepsilon\}$ is empty, the same conclusion is immediate. Thus, (17) is satisfied.
Since φ is continuous and equals c on the compact set A, uniform continuity combined with (17) yields

$$\frac{1}{|A_m|} \sum_{u\in A_m} \varphi(u) \xrightarrow{m\to\infty} c,$$

which completes the proof. □

Lemma 3.3. *Let (U_n, V_n) be jointly Gaussian random elements of $C(E)\times C(F)$, with marginal laws independent of n. If*

$$\sup_{u\in E,\, v\in F} |\operatorname{\mathbb{C}ov}(U_n(u), V_n(v))| \xrightarrow{n\to\infty} 0,$$

then

$$(U_n, V_n) \xrightarrow[n\to\infty]{d} (U, V) \quad \text{in } C(E)\times C(F),$$

where U and V are independent random elements with these marginal laws.

Proof. Tightness follows from the fixed marginal laws. For every finite collection of points, the corresponding Gaussian vectors have fixed marginal covariance blocks and off-diagonal blocks tending to zero. Hence their finite-dimensional distributions converge to product Gaussian distributions. Every subsequential weak limit therefore has the product finite-dimensional distributions, which determine the product Gaussian law on $C(E)\times C(F)$. □

3.2. **Covariance Price tools.** Before stating the Price-interpolation tools needed to study the workload covariance asymptotics, we introduce notation and measurability conventions used below. For every non-empty compact metric space K, we introduce a Borel map

$$\mathfrak{a}_K : C(K) \longrightarrow K, \qquad \mathfrak{a}_K(h) \in \arg\max_{u\in K} h(u).$$

Such a selector exists by [3, Theorem 18.19]. Clearly, if $(\theta,\omega)\mapsto Z^{(\theta)}(\omega)$ is a jointly measurable $C(K)$-valued field, then $(\theta,\omega)\mapsto \mathfrak{a}_K(Z^{(\theta)}(\omega))$ is jointly measurable.
On $K=[0,T]$, we take $\mathfrak{a}_K$ to be the leftmost maximiser, and for a continuous path h on $[0,\infty)$ satisfying $h(t)\xrightarrow{t\to\infty}-\infty$, we define $\mathfrak{a}_\infty(h)$ to be its leftmost global maximiser. This map is measurable in the topology of uniform convergence on compact sets, since

$$\mathfrak{a}_\infty(h) = \lim_{n\to\infty} \mathfrak{a}_{[0,n]}(h|_{[0,n]}),$$

and the sequence is eventually constant once the compact interval contains the global maximising set.

Lemma 3.4. *If (X,Y) is a jointly Gaussian element of $C(E)\times C(F)$ with cross-covariance*

$$R(u,v) := \operatorname{\mathbb{C}ov}(X(u), Y(v)), \qquad u\in E,\ v\in F,$$

then almost surely R is constant on

$$\arg\max_E X \times \arg\max_F Y.$$

Consequently, its value is independent of the measurable maximiser selections.

Proof. We first recall the Cameron–Martin tie observation. Let Z be a continuous Gaussian process on a compact metric space K, with the reproducing kernel Hilbert space (RKHS) $\mathcal{H}_Z$. We shall show that every fixed $h\in\mathcal{H}_Z$ is almost surely constant on $\arg\max_K Z$. Indeed, for $z\in C(K)$, let

$$\phi_z(r) := \max_{t\in K}\{z(t)+rh(t)\}.$$

This is a convex Lipschitz function of r. The one-dimensional Danskin formulae (see, e.g., [4, Proposition A.3.2]) for the right and left derivatives of ϕ_z give

$$\phi'_{z,+}(r) = \max_{t\in\arg\max(z+rh)} h(t), \qquad \phi'_{z,-}(r) = \min_{t\in\arg\max(z+rh)} h(t).$$

Thus h is nonconstant on $\arg\max(z+rh)$ precisely when ϕ_z is not differentiable at r. For each z, this occurs at most at countably many r. The bad set in $C(K)\times\mathbb{R}$ is Borel, because the one-sided derivatives are limits of rational difference quotients. Hence, by Fubini theorem there exists a value r for which the bad event has probability zero for $Z+rh$. Due to Cameron–Martin theorem [6, Section 2.4], the laws of $Z+rh$ and Z are equivalent. Therefore h is almost surely constant on $\arg\max_K Z$.
Let $\mathcal{H}_X$ be the RKHS of X. Since for each fixed $v\in F$, $R(\cdot,v)$ belongs to $\mathcal{H}_X$, then the tie observation shows that $R(\cdot,v)$ is constant on $\arg\max_E X$ almost surely. Intersecting these events over a countable dense subset of F and then using continuity of R yields the assertion simultaneously for every $v\in F$. The symmetric argument with the RKHS of Y confirms that $R(u,\cdot)$ is constant on $\arg\max_F Y$, simultaneously for every $u\in E$. Combining the two statements completes the proof. □

For a given jointly Gaussian element (X, Y) of $C(E) \times C(F)$ we set $m_X(u) := \mathbb{E}\{X(u)\}$, $m_Y(v) := \mathbb{E}\{Y(v)\}$ and $R(u, v) := \mathbb{C}\mathrm{ov}(X(u), Y(v))$. Let X' and Y' be independent marginal copies of X and Y that are independent of (X, Y), and

$$X^{(\theta)} := m_X + \sqrt{\theta}\,(X - m_X) + \sqrt{1-\theta}\,(X' - m_X),$$
$$Y^{(\theta)} := m_Y + \sqrt{\theta}\,(Y - m_Y) + \sqrt{1-\theta}\,(Y' - m_Y)$$

for $\theta \in [0, 1]$. Moreover, let

$$M_X := \max_{u\in E} X(u), \qquad M_Y := \max_{v\in F} Y(v),$$

and let τ_X, τ_Y denote the corresponding maximisers.

Lemma 3.5. *If (X, Y) is a jointly Gaussian element of $C(E) \times C(F)$, then*

$$\mathbb{C}\mathrm{ov}(M_X, M_Y) = \int_0^1 \mathbb{E}\{R(\tau_{X^{(\theta)}}, \tau_{Y^{(\theta)}})\}\, d\theta.$$

Proof. Let us choose finite nets $D_m \subset E$ and $F_m \subset F$ with mesh tending to zero. For $\gamma > 0$, define

$$M^X_{m,\gamma} := \frac{1}{\gamma} \log \sum_{u\in D_m} e^{\gamma X(u)}, \qquad M^Y_{m,\gamma} := \frac{1}{\gamma} \log \sum_{v\in F_m} e^{\gamma Y(v)}.$$

Apply Lemma 3.1 to the centred finite-dimensional vectors $(X(u) - m_X(u))_{u\in D_m}$ and $(Y(v) - m_Y(v))_{v\in F_m}$. In these centred coordinates, the two functions are

$$z \longmapsto \frac{1}{\gamma} \log \sum_{u\in D_m} e^{\gamma(m_X(u)+z_u)}, \qquad w \longmapsto \frac{1}{\gamma} \log \sum_{v\in F_m} e^{\gamma(m_Y(v)+w_v)}.$$

Their derivatives are the softmax weights:

$$\partial_{X(u)} M^X_{m,\gamma} = \frac{e^{\gamma X(u)}}{\sum_{z\in D_m} e^{\gamma X(z)}}, \qquad u \in D_m,$$

and analogously for the Y element. Applying Lemma 3.1 therefore yields

$$\mathbb{C}\mathrm{ov}(M^X_{m,\gamma}, M^Y_{m,\gamma}) = \int_0^1 \mathbb{E}\left\{ \int_E \int_F R(u, v)\pi^{X^{(\theta)}}_{m,\gamma}(du)\pi^{Y^{(\theta)}}_{m,\gamma}(dv) \right\} d\theta. \tag{18}$$

For every $\theta \in (0, 1]$, $\mathbb{C}\mathrm{ov}(X^{(\theta)}(u), Y^{(\theta)}(v)) = \theta R(u, v)$. In view of Lemma 3.4, applied to $(X^{(\theta)}, Y^{(\theta)})$, the kernel θR, and hence R, is constant on

$$\arg\max_E X^{(\theta)} \times \arg\max_F Y^{(\theta)} \tag{19}$$

almost surely. On the product net $D_m \times F_m$, the product measure $\pi^{X^{(\theta)}}_{m,\gamma} \otimes \pi^{Y^{(\theta)}}_{m,\gamma}$ is the softmax measure associated with the function $(u, v) \mapsto X^{(\theta)}(u) + Y^{(\theta)}(v)$, whose maximising set is the Cartesian product (19). Hence, Lemma 3.2 yields the following convergence as $\gamma \to \infty$ and then $m \to \infty$

$$\lim_{\gamma\to\infty} \int_E \int_F R(u, v)\pi^{X^{(\theta)}}_{m,\gamma}(du)\pi^{Y^{(\theta)}}_{m,\gamma}(dv) \xrightarrow{m\to\infty} R(\tau_{X^{(\theta)}}, \tau_{Y^{(\theta)}})$$

for $d\theta \otimes d\mathbb{P}$-almost every (θ, ω).
We note that the integrand on the right-hand side of (18) is bounded in absolute value by $\|R\|_\infty$. Thus, dominated convergence on $[0, 1] \times \Omega$ justifies passing first to the limit $\gamma \to \infty$ and then to the limit $m \to \infty$.
On the left-hand side, for each fixed m,

$$0 \le M^X_{m,\gamma} - \max_{u\in D_m} X(u) \le \frac{\log |D_m|}{\gamma},$$

so the soft maximum converges to the finite-net maximum in L^2 as $\gamma \to \infty$. As $m \to \infty$, pathwise uniform continuity gives

$$\max_{u\in D_m} X(u) \longrightarrow M_X \qquad \text{almost surely.}$$

Since these maxima are bounded in absolute value by $\|X\|_\infty$, which has a finite second moment, this convergence also holds in L^2. The same arguments apply to Y. Consequently, the covariance on the left-hand side converges to $\mathbb{C}\mathrm{ov}(M_X, M_Y)$ under these successive limits.
This completes the proof. □

Lemma 3.6. *Let (X_r, Y_r) be jointly Gaussian in $C(E)\times C(F)$ with marginal laws independent of r. Assume that there exist continuous functions $g_1 : E \to \mathbb{R}$ and $g_2 : F \to \mathbb{R}$ and a function $b(r) > 0$ such that $b(r) \xrightarrow{r\to\infty} 0$ and*

$$\sup_{u\in E,\, v\in F} \left| \frac{\mathbb{C}\mathrm{ov}(X_r(u), Y_r(v))}{b(r)} - g_1(u)g_2(v) \right| \xrightarrow{r\to\infty} 0.$$

Let $\widetilde{X}$ and $\widetilde{Y}$ be independent random elements with the marginal laws of X_r and Y_r, respectively. If the marginal maximisers are a.s. unique, then

$$\frac{\mathbb{C}\mathrm{ov}(M_{X_r}, M_{Y_r})}{b(r)} \xrightarrow{r\to\infty} \mathbb{E}\{g_1(\tau_{\widetilde{X}})\}\,\mathbb{E}\{g_2(\tau_{\widetilde{Y}})\}.$$

Proof. Let $R_r(u,v) := \mathbb{C}\mathrm{ov}(X_r(u), Y_r(v))$ and $K_r(u,v) := \frac{R_r(u,v)}{b(r)}$. In light of Lemma 3.5 we have

$$\frac{\mathbb{C}\mathrm{ov}(M_{X_r}, M_{Y_r})}{b(r)} = \int_0^1 \mathbb{E}\Big\{K_r(\tau_{X_r^{(\theta)}}, \tau_{Y_r^{(\theta)}})\Big\}\, d\theta.$$

Uniform convergence allows us to replace K_r by $g_1 g_2$ at an error tending to zero as $r\to\infty$. Moreover we have

$$\mathbb{C}\mathrm{ov}(X_r^{(\theta)}(u), Y_r^{(\theta)}(v)) = \theta R_r(u,v).$$

Since $b(r) \xrightarrow{r\to\infty} 0$, the above covariance tends to zero uniformly on $E\times F$. Hence Lemma 3.3 implies decoupling of the interpolated compact pair for each fixed θ. The map $x \mapsto (M_x, \tau_x)$ is continuous at every continuous path with a unique maximiser. Since g_1 and g_2 are continuous and bounded on their compact domains, we obtain

$$\mathbb{E}\Big\{g_1(\tau_{X_r^{(\theta)}})g_2(\tau_{Y_r^{(\theta)}})\Big\} \xrightarrow{r\to\infty} \mathbb{E}\big\{g_1(\tau_{\widetilde{X}})g_2(\tau_{\widetilde{Y}})\big\} = \mathbb{E}\{g_1(\tau_{\widetilde{X}})\}\,\mathbb{E}\{g_2(\tau_{\widetilde{Y}})\}.$$

The limit is independent of θ, so dominated convergence on $[0,1]$ establishes the claim. □

We now fix the notation for the half-line interpolation results below. For each $r \ge 0$ let $X_r(t),\ t \ge 0$ and $Y_r(t),\ t \ge 0$ be jointly Gaussian real-valued processes with continuous sample paths a.s., whose marginal laws do not depend on r. Let $\widetilde{X}$ and $\widetilde{Y}$ be independent processes with these marginal laws, and put $m_X(t) := \mathbb{E}\{X_r(t)\}$, $m_Y(t) := \mathbb{E}\{Y_r(t)\}$ for their continuous means. Let X_r' and Y_r' be independent marginal copies of X_r and Y_r, respectively, independent of (X_r, Y_r). Set

$$R_r(u,v) := \mathbb{C}\mathrm{ov}(X_r(u), Y_r(v)), \qquad u, v \ge 0,$$

and for $\theta \in [0,1]$ put

$$X_r^{(\theta)} := m_X + \sqrt{\theta}\,(X_r - m_X) + \sqrt{1-\theta}\,(X_r' - m_X),$$
$$Y_r^{(\theta)} := m_Y + \sqrt{\theta}\,(Y_r - m_Y) + \sqrt{1-\theta}\,(Y_r' - m_Y).$$

Also let $b(r) > 0$.

Lemma 3.7. *Suppose that the processes $\widetilde{X}$ and $\widetilde{Y}$ tend to $-\infty$ at infinity a.s. and have global maxima with finite second moments. Assume that there are nonnegative nondecreasing functions G_X, G_Y such that for every $r \ge 0$ we have*

$$|R_r(u,v)| \le G_X(u)G_Y(v), \qquad u, v \ge 0, \tag{20}$$

and for each fixed r

$$\sup_{\theta\in[0,1]} \mathbb{E}\Big\{G_X(\tau_{X_r^{(\theta)}})G_Y(\tau_{Y_r^{(\theta)}})\Big\} < \infty. \tag{21}$$

Then,

$$\mathbb{C}\mathrm{ov}(M_{X_r}, M_{Y_r}) = \int_0^1 \mathbb{E}\Big\{R_r(\tau_{X_r^{(\theta)}}, \tau_{Y_r^{(\theta)}})\Big\}\, d\theta. \tag{22}$$

If in addition

$$\lim_{T\to\infty}\limsup_{r\to\infty}\sup_{\theta\in[0,1]}\frac{1}{b(r)}\mathbb{E}\left\{\begin{array}{l}\left|R_r(\tau_{X_r^{(\theta)}},\tau_{Y_r^{(\theta)}})-R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}})\right|\\ \quad+\left|R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}})-R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}}(T))\right|\end{array}\right\}=0, \tag{23}$$

then

$$\lim_{T\to\infty}\limsup_{r\to\infty}\left|\frac{\mathbb{C}\mathrm{ov}(M_{X_r},M_{Y_r})}{b(r)}-\frac{\mathbb{C}\mathrm{ov}(M_{X_r}(T),M_{Y_r}(T))}{b(r)}\right|=0. \tag{24}$$

Proof. We note that if x is a continuous path tending to $-\infty$ and its compact and global maximising locations are chosen to be the leftmost ones, then for every nonnegative nondecreasing w we have

$$\big(w(\tau_x(T))+w(\tau_x)\big)\mathbf{1}_{\{\tau_x(T)\neq\tau_x\}}\le 2w(\tau_x)\mathbf{1}_{\{\tau_x>T\}}. \tag{25}$$

Indeed, on $\{\tau_x\le T\}$ the leftmost compact and global maximisers agree. On $\{\tau_x>T\}$, one has $\tau_x(T)\le T<\tau_x$, and monotonicity of w yields the bound.
Fix r and let $A<\infty$. Applying Lemma 3.5 on $[0,A]^2$ yields

$$\mathbb{C}\mathrm{ov}(M_{X_r}(A),M_{Y_r}(A))=\int_0^1\mathbb{E}\Big\{R_r(\tau_{X_r^{(\theta)}}(A),\tau_{Y_r^{(\theta)}}(A))\Big\}\,d\theta. \tag{26}$$

Because the processes tend to $-\infty$, $M_{X_r}(A)\xrightarrow{A\to\infty}M_{X_r}$ and $M_{Y_r}(A)\xrightarrow{A\to\infty}M_{Y_r}$ a.s. Moreover we have

$$|M_{X_r}(A)|\le|X_r(0)|+|M_{X_r}|,\qquad |M_{Y_r}(A)|\le|Y_r(0)|+|M_{Y_r}|.$$

The global maxima have finite second moments, so the left-hand side of (26) converges to $\mathbb{C}\mathrm{ov}(M_{X_r},M_{Y_r})$. Using the kernel bound (20), monotonicity of G_X,G_Y and (25) we obtain

$$\left|R_r(\tau_{X_r^{(\theta)}}(A),\tau_{Y_r^{(\theta)}}(A))-R_r(\tau_{X_r^{(\theta)}},\tau_{Y_r^{(\theta)}})\right|\le CG_X(\tau_{X_r^{(\theta)}})G_Y(\tau_{Y_r^{(\theta)}})\left(\mathbf{1}_{\{\tau_{X_r^{(\theta)}}>A\}}+\mathbf{1}_{\{\tau_{Y_r^{(\theta)}}>A\}}\right).$$

Let us observe, that for every fixed θ, the displayed bound tends to zero a.s. as $A\to\infty$, and hence it does so for $d\theta\otimes d\mathbb{P}$-almost every (θ,ω). Moreover it is bounded by

$$C\,G_X(\tau_{X_r^{(\theta)}})G_Y(\tau_{Y_r^{(\theta)}}),$$

whose integral over $[0,1]\times\Omega$ is at most

$$C\sup_{\theta\in[0,1]}\mathbb{E}\Big\{G_X(\tau_{X_r^{(\theta)}})G_Y(\tau_{Y_r^{(\theta)}})\Big\}<\infty.$$

Thus, by the dominated convergence on $[0,1]\times\Omega$, equality (22) holds.
To establish (24), we apply (22) and (26) with $A=T$, and divide both identities by $b(r)$. Specifically,

$$\begin{aligned}&\frac{\mathbb{C}\mathrm{ov}(M_{X_r},M_{Y_r})}{b(r)}-\frac{\mathbb{C}\mathrm{ov}(M_{X_r}(T),M_{Y_r}(T))}{b(r)}\\&\quad=\frac{1}{b(r)}\int_0^1\mathbb{E}\Big\{R_r(\tau_{X_r^{(\theta)}},\tau_{Y_r^{(\theta)}})-R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}}(T))\Big\}\,d\theta.\end{aligned}$$

Adding and subtracting $R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}})$ yields

$$\begin{aligned}R_r(\tau_{X_r^{(\theta)}},\tau_{Y_r^{(\theta)}})-R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}}(T))=&\big[R_r(\tau_{X_r^{(\theta)}},\tau_{Y_r^{(\theta)}})-R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}})\big]\\&+\big[R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}})-R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}}(T))\big].\end{aligned}$$

Consequently, by the triangle inequality,

$$\begin{aligned}&\left|\frac{\mathbb{C}\mathrm{ov}(M_{X_r},M_{Y_r})}{b(r)}-\frac{\mathbb{C}\mathrm{ov}(M_{X_r}(T),M_{Y_r}(T))}{b(r)}\right|\\&\le\sup_{\theta\in[0,1]}\frac{1}{b(r)}\mathbb{E}\left\{\begin{array}{l}\left|R_r(\tau_{X_r^{(\theta)}},\tau_{Y_r^{(\theta)}})-R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}})\right|\\ \quad+\left|R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}})-R_r(\tau_{X_r^{(\theta)}}(T),\tau_{Y_r^{(\theta)}}(T))\right|\end{array}\right\}.\end{aligned}$$

Taking $\limsup_{r\to\infty}$ on both sides and then letting $T\to\infty$, the right-hand side tends to zero by (23), which proves (24). □

Theorem 3.8. *Under the assumptions and with the notation of Lemma 3.7, let $b(r)>0$ with $b(r)\xrightarrow{r\to\infty}0$ and assume that (23) holds. Assume that for each fixed $T>0$ the maximisers of $\widetilde{X}$ and $\widetilde{Y}$ over $[0,T]$ are almost surely unique. If there exist continuous functions $g_1,g_2:[0,\infty)\to\mathbb{R}$ such that for every $T>0$*

$$\sup_{0\le u,v\le T}\left|\frac{R_r(u,v)}{b(r)}-g_1(u)g_2(v)\right|\xrightarrow{r\to\infty}0 \tag{27}$$

and

$$g_1(\tau_{\widetilde{X}}(T))\xrightarrow[T\to\infty]{L^1} g_1(\tau_{\widetilde{X}}),\qquad g_2(\tau_{\widetilde{Y}}(T))\xrightarrow[T\to\infty]{L^1} g_2(\tau_{\widetilde{Y}}), \tag{28}$$

then

$$\lim_{r\to\infty}\frac{\mathbb{C}\mathrm{ov}(M_{X_r},M_{Y_r})}{b(r)}=\mathbb{E}\{g_1(\tau_{\widetilde{X}})\}\,\mathbb{E}\{g_2(\tau_{\widetilde{Y}})\}. \tag{29}$$

Proof. Fix $T>0$. In view of Lemma 3.6 and (27) we have

$$\lim_{r\to\infty}\frac{\mathbb{C}\mathrm{ov}(M_{X_r}(T),M_{Y_r}(T))}{b(r)}=\mathbb{E}\{g_1(\tau_{\widetilde{X}}(T))\}\,\mathbb{E}\{g_2(\tau_{\widetilde{Y}}(T))\}.$$

Applying Lemma 3.7 we obtain

$$\lim_{T\to\infty}\limsup_{r\to\infty}\left|\frac{\mathbb{C}\mathrm{ov}(M_{X_r},M_{Y_r})}{b(r)}-\frac{\mathbb{C}\mathrm{ov}(M_{X_r}(T),M_{Y_r}(T))}{b(r)}\right|=0,$$

hence for every $T>0$ we have

$$\begin{aligned}&\limsup_{r\to\infty}\left|\frac{\mathbb{C}\mathrm{ov}(M_{X_r},M_{Y_r})}{b(r)}-\mathbb{E}\{g_1(\tau_{\widetilde{X}})\}\,\mathbb{E}\{g_2(\tau_{\widetilde{Y}})\}\right|\\&\quad\le\limsup_{r\to\infty}\left|\frac{\mathbb{C}\mathrm{ov}(M_{X_r},M_{Y_r})}{b(r)}-\frac{\mathbb{C}\mathrm{ov}(M_{X_r}(T),M_{Y_r}(T))}{b(r)}\right|\\&\qquad+\left|\mathbb{E}\{g_1(\tau_{\widetilde{X}}(T))\}\,\mathbb{E}\{g_2(\tau_{\widetilde{Y}}(T))\}-\mathbb{E}\{g_1(\tau_{\widetilde{X}})\}\,\mathbb{E}\{g_2(\tau_{\widetilde{Y}})\}\right|.\end{aligned}$$

Letting $T\to\infty$ and using (28), we arrive at (29). □

Remark 3.9. The preceding theorem can be extended to a continuous limiting function $\Gamma(u,v)$ in place of $g_1(u)g_2(v)$. Specifically, if we assume that $R_r/b(r)\to\Gamma$ locally uniformly, retain condition (23), and

$$\Gamma(\tau_{\widetilde{X}}(T),\tau_{\widetilde{Y}}(T))\xrightarrow[T\to\infty]{L^1}\Gamma(\tau_{\widetilde{X}},\tau_{\widetilde{Y}}),$$

them, the same proof gives the limit $\mathbb{E}\{\Gamma(\tau_{\widetilde{X}},\tau_{\widetilde{Y}})\}$ in (29), where $\widetilde{X}$ and $\widetilde{Y}$ are independent processes with the prescribed marginal laws. We do not formulate this additional generality separately, since the models treated here have explicit product kernels.

3.3. **Uniqueness and tail estimates.**

Lemma 3.10. *Let $\mathbb{T}$ be a σ-compact metric space, and let $Z(t)$, $t\in\mathbb{T}$, be a Gaussian process with continuous sample paths. If*

$$\mathbb{V}\mathrm{ar}(Z(s)-Z(t))>0,\qquad s,t\in\mathbb{T},\quad s\neq t,$$

then almost surely Z cannot attain its supremum at two distinct points.

Proof. Let $\mathbb{T}=\bigcup_{n\ge1}K_n$, where the sets K_n are compact and increasing. By [13, Lemma 2.6], almost surely Z has a unique maximiser on each K_n. If two distinct points attained the supremum over $\mathbb{T}$, then both would belong to some K_n, contradicting uniqueness on that compact set. □

Lemma 3.11. *Let X be a centred Gaussian process with stationary increments, continuous sample paths and $X(0) = 0$ almost surely. Suppose that its variance function V_X satisfies*

$$\frac{V_X(t)\log t}{t^2} \xrightarrow{t\to\infty} 0, \tag{30}$$

and $V_X(t) > 0$ for every $t > 0$. For every $c > 0$, put $W(t) = X(t) - ct$. Then $W(t) \xrightarrow{t\to\infty} -\infty$ almost surely, its maximum over $[0,\infty)$ is finite and attained, and its compact and global maximising locations are almost surely unique. Furthermore, $M_W, \tau_W \in L^p$ for every $p > 0$, and, for every $N > 0$,

$$\mathbb{P}\{\tau_W > T\} = O(T^{-N}), \qquad T \to \infty.$$

If, in addition, $V_X(t)/t \xrightarrow{t\to\infty} D_X \in [0,\infty)$, then

$$\mathbb{E}\{e^{q\tau_W}\} < \infty, \qquad 0 < q < \frac{c^2}{2D_X}, \tag{31}$$

where $c^2/(2D_X)$ is interpreted as $+\infty$ when $D_X = 0$.

Proof. Set

$$U_X(x) := \sup_{0\le t\le x} V_X(t), \qquad S_n := \sup_{n\le t\le n+1} X(t), \qquad m_* := \mathbb{E}\{\sup_{0\le t\le 1} X(t)\}, \qquad n \ge 0.$$

Continuity and Fernique's theorem imply $m_* < \infty$; see [6]. Since

$$S_n = X(n) + \sup_{0\le h\le 1}(X(n+h) - X(n)),$$

stationary increments yield $\mathbb{E}\{S_n\} = m_*$. Put $v_n := \sup_{n\le t\le n+1} V_X(t)$. Borell–TIS [2] yields

$$\mathbb{P}\{S_n \ge x\} \le \exp\left(-\frac{(x-m_*)_+^2}{2v_n}\right), \qquad x \in \mathbb{R}. \tag{32}$$

Condition (30) implies $U_X(n+1) = o(n^2/\log n)$, and $v_n \le U_X(n+1)$. Hence, for every $a > 0$ and every $N > 0$,

$$\mathbb{P}\{S_n \ge an\} = O(n^{-N}), \qquad n \to \infty.$$

Similarly, the same estimate applies to the supremum of $-X$. Hence, Borel–Cantelli lemma, applied to positive rational a, therefore yields $X(t)/t \xrightarrow{t\to\infty} 0$ almost surely. Thus $W(t) \xrightarrow{t\to\infty} -\infty$, and continuity implies finiteness and attainment of M_W. The inequality

$$\mathbb{V}\mathrm{ar}(W(s) - W(t)) = V_X(|s-t|) > 0, \qquad s \ne t,$$

and Lemma 3.10 yield uniqueness on every fixed compact interval. Almost surely, the maximiser is unique on every interval $[0,n]$, $n \in \mathbb{N}$, simultaneously. Any two global maximisers would lie in one such interval, so the global maximiser is also almost surely unique.
Because $W(0) = 0$, the event $\{\tau_W > T\}$ implies $\sup_{t\ge T} W(t) \ge 0$. For all sufficiently large T,

$$\mathbb{P}\{\tau_W > T\} \le \sum_{n\ge\lfloor T\rfloor} \mathbb{P}\{S_n \ge cn\} = O(T^{-N})$$

for every $N > 0$, where we used the estimate $\mathbb{P}\{S_n \ge cn\} = O(n^{-(N+2)})$ and summed over $n \ge \lfloor T\rfloor$. Thus τ_W has finite moments of every positive order.
We next show that $\mathbb{E}\{M_W^p\} < \infty$ for every $p > 0$. Let $p > 0$ be arbitrary. For all sufficiently large n, $a_n := cn - m_* \ge cn/2$, and (32) yields

$$\begin{aligned}
\mathbb{E}\{(S_n - cn)_+^p\} &= p\int_0^\infty z^{p-1}\mathbb{P}\{S_n > cn + z\}\,dz \\
&\le pe^{-a_n^2/(4v_n)}\int_0^\infty z^{p-1}e^{-z^2/(4v_n)}\,dz \\
&\le C_p v_n^{p/2}e^{-c^2n^2/(16v_n)}.
\end{aligned}$$

Here we used $(a_n+z)^2 \geq (a_n^2+z^2)/2$. The last bound is summable, since $v_n = o(n^2/\log n)$. For each of the finitely many remaining intervals $[n, n+1]$, the random variable $\sup_{n\leq t\leq n+1}|X(t)|$ has finite moments of every positive order by Fernique's theorem. Since

$$0 \leq M_W^p \leq \sum_{n=0}^{\infty}(S_n - cn)_+^p,$$

Tonelli's theorem implies $M_W \in L^p$.

Finally, suppose $V_X(t)/t \xrightarrow{t\to\infty} D_X$. Then $v_n = D_X n + o(n)$. If $D_X > 0$, for every $a \in (0, c^2/(2D_X))$, (32) implies $\mathbb{P}\{S_n \geq cn\} \leq e^{-an}$ for all large n. If $D_X = 0$, the same estimate holds for every $a > 0$. Hence, summing over $n \geq \lfloor T \rfloor$ yields $\mathbb{P}\{\tau_W > T\} \leq C_a e^{-aT}$. Thus, for any $q < a$,

$$\mathbb{E}\{e^{q\tau_W}\} = 1 + q\int_0^{\infty} e^{qt}\mathbb{P}\{\tau_W > t\}\,dt < \infty.$$

Choosing $a > q$ proves (31). □

4. Proofs

For the non-Markovian proofs we use the following notation. For the relevant input process X we set

$$\mathcal{X}(u) := X(0) - X(-u) - c_1 u, \qquad \mathcal{Y}_r(v) := X(r) - X(r-v) - c_2 v, \qquad u, v \geq 0. \tag{33}$$

By definition we have

$$Q_{W_1}(0) = M_{\mathcal{X}}, \qquad Q_{W_2}(r) = M_{\mathcal{Y}_r},$$

and the stationarity of increments implies that the marginal laws of $\mathcal{X}$ and $\mathcal{Y}_r$ are those of W_1 and W_2, respectively.

4.1. **Proof of Theorem 2.1.** First we note that the pair (Q_{W_1}, Q_{W_2}) is jointly stationary. Let $\mathcal{F}_0 := \sigma\{B(s) : s \leq 0\}$. The random variables $Q_{W_1}(0)$ and $Q_{W_2}(0)$ are $\mathcal{F}_0$-measurable while the future increments $B(u) - B(0),\ u \geq 0$ are independent of $\mathcal{F}_0$.

Proof of (6). Recalling $m_1 = \mathbb{E}\{Q_{W_1}(0)\} = 1/(2c_1)$, we note that

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0), Q_{W_2}(r)\big) = \mathbb{E}\{(Q_{W_1}(0) - m_1)Q_{W_2}(r)\} = \mathbb{E}\{(Q_{W_1}(0) - m_1)\mathbb{E}\{Q_{W_2}(r) \mid \mathcal{F}_0\}\}. \tag{34}$$

We first identify the conditional dynamics of $Q_{W_2}(r)$ for $r \geq 0$. For $x \geq 0$, define

$$R_{W_2}^x(t) := \Big(-\inf_{0\leq u\leq t}\{W_2(u) + x\}\Big)^+, \qquad Q_{W_2}^x(t) := W_2(t) + x + R_{W_2}^x(t).$$

Clearly, $Q_{W_2}^x$ is reflected Brownian motion with drift $-c_2$, started from x and driven by the future Brownian increments. Moreover, conditionally on $\mathcal{F}_0$, the process $Q_{W_2}(r),\ r \geq 0$ has the law of reflected Brownian motion with drift $-c_2$, started from $Q_{W_2}(0)$ and driven by the future Brownian increments.

For $a \geq 0$ define

$$H_r^{(2)}(a) := \mathbb{P}\{T_a^{(c_2)} \leq r\}.$$

We have

$$\mathbb{E}\{R_{W_2}^x(r)\} = \int_0^{\infty}\mathbb{P}\{R_{W_2}^x(r) > y\}\,dy = \int_0^{\infty} H_r^{(2)}(x+y)\,dy,$$

and therefore

$$\mathbb{E}\{Q_{W_2}^x(r)\} = x - c_2 r + \int_0^{\infty} H_r^{(2)}(x+y)\,dy = x - c_2 r + \int_x^{\infty} H_r^{(2)}(a)\,da,$$

implying

$$\mathbb{E}\{Q_{W_2}(r) \mid \mathcal{F}_0\} = Q_{W_2}(0) - c_2 r + \int_{Q_{W_2}(0)}^{\infty} H_r^{(2)}(a)\,da.$$

Combining (34) with $\mathbb{E}\{Q_{W_1}(0) - m_1\} = 0$ yields

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0), Q_{W_2}(r)\big) = \mathbb{E}\{(Q_{W_1}(0) - m_1)Q_{W_2}(0)\} + \mathbb{E}\left\{(Q_{W_1}(0) - m_1)\int_{Q_{W_2}(0)}^{\infty} H_r^{(2)}(a)\,da\right\}.$$

By Fubini's theorem we have

$$\mathbb{E}\Big\{(Q_{W_1}(0)-m_1)\int_{Q_{W_2}(0)}^{\infty} H_r^{(2)}(a)\,da\Big\} = \int_0^{\infty} H_r^{(2)}(a)\mathbb{E}\Big\{(Q_{W_1}(0)-m_1)\mathbf{1}_{\{Q_{W_2}(0)\le a\}}\Big\}\,da.$$

For $a \ge 0$ define

$$L_{12}(a) := \mathbb{E}\Big\{(Q_{W_1}(0)-m_1)\mathbf{1}_{\{Q_{W_2}(0)>a\}}\Big\}.$$

Since $\mathbb{E}\{Q_{W_1}(0)-m_1\}=0$ we have

$$\mathbb{E}\Big\{(Q_{W_1}(0)-m_1)\mathbf{1}_{\{Q_{W_2}(0)\le a\}}\Big\} = -L_{12}(a).$$

Moreover,

$$Q_{W_2}(0) = \int_0^{\infty} \mathbf{1}_{\{Q_{W_2}(0)>a\}}\,da,$$

and

$$\int_0^{\infty} |L_{12}(a)|\,da \le \mathbb{E}\{|Q_{W_1}(0)-m_1|Q_{W_2}(0)\} < \infty.$$

Another application of Fubini yields

$$\mathbb{E}\{(Q_{W_1}(0)-m_1)Q_{W_2}(0)\} = \int_0^{\infty} L_{12}(a)\,da.$$

Combining these identities, we get

$$\begin{aligned}\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big) &= \int_0^{\infty} L_{12}(a)\,da - \int_0^{\infty} H_r^{(2)}(a)L_{12}(a)\,da\\ &= \int_0^{\infty} L_{12}(a)\{1-H_r^{(2)}(a)\}\,da\\ &= \int_0^{\infty} \mathbb{E}\Big\{(Q_{W_1}(0)-m_1)\mathbf{1}_{\{Q_{W_2}(0)>a\}}\Big\}\mathbb{P}\{T_a^{(c_2)}>r\}\,da,\end{aligned}$$

which is the covariance identity (6).

Proof of (7). For $a>0$ the hitting time $T_a^{(c_2)}$ has density (see [7])

$$f_a^{(2)}(u) = \frac{a}{\sqrt{2\pi}u^{3/2}}\exp\left(-\frac{(a-c_2u)^2}{2u}\right), \qquad u>0.$$

Since

$$\int_0^{\infty} |L_{12}(a)|\mathbb{P}\{T_a^{(c_2)}>r\}\,da \le \int_0^{\infty}|L_{12}(a)|\,da<\infty,$$

we may substitute the density and interchange the order of integration in (6) to obtain

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big) = \int_0^{\infty} L_{12}(a)\int_r^{\infty}\frac{a}{\sqrt{2\pi}u^{3/2}}\exp\left(-\frac{(a-c_2u)^2}{2u}\right)du\,da.$$

Using

$$-\frac{(a-c_2u)^2}{2u} = -\frac{c_2^2u}{2}+c_2a-\frac{a^2}{2u},$$

and defining

$$J_{12}(u) := \int_0^{\infty} ae^{c_2a}e^{-a^2/(2u)}L_{12}(a)\,da, \qquad u>0,$$

we obtain

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big) = \frac{1}{\sqrt{2\pi}}\int_r^{\infty} u^{-3/2}e^{-c_2^2u/2}J_{12}(u)\,du.$$

We next prove that $J_{12}(u)$ has a finite limit as $u\to\infty$. Since

$$|L_{12}(a)| \le \mathbb{E}\Big\{|Q_{W_1}(0)-m_1|\mathbf{1}_{\{Q_{W_2}(0)>a\}}\Big\},$$

we have

$$\int_0^\infty ae^{c_2 a}|L_{12}(a)|\,da \le \mathbb{E}\left\{|Q_{W_1}(0)-m_1|\int_0^{Q_{W_2}(0)} ae^{c_2 a}\,da\right\} \le C\mathbb{E}\Big\{|Q_{W_1}(0)-m_1|(1+Q_{W_2}(0))e^{c_2 Q_{W_2}(0)}\Big\} < \infty,$$

for a finite constant C depending only on c_2. For $q\in(1,2)$ and $p=q/(q-1)$, Hölder's inequality yields

$$\mathbb{E}\Big\{|Q_{W_1}(0)-m_1|(1+Q_{W_2}(0))e^{c_2 Q_{W_2}(0)}\Big\} \le (\mathbb{E}\{|Q_{W_1}(0)-m_1|^p\})^{\frac{1}{p}}\left(\mathbb{E}\{(1+Q_{W_2}(0))^q e^{qc_2 Q_{W_2}(0)}\}\right)^{\frac{1}{q}} \tag{35}$$

where the right-hand side is finite because $Q_{W_1}(0)\sim \mathrm{Exp}(2c_1)$, $Q_{W_2}(0)\sim\mathrm{Exp}(2c_2)$ and $qc_2<2c_2$. Therefore

$$ae^{c_2 a}L_{12}(a)\in L^1(0,\infty).$$

Hence, the dominated convergence yields

$$J_{12}(u)\xrightarrow{u\to\infty} J_{12}(\infty):=\int_0^\infty ae^{c_2 a}L_{12}(a)\,da.$$

Let

$$A_2(r):=\int_r^\infty u^{-3/2}e^{-c_2^2 u/2}\,du.$$

We can write

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big)=\frac{J_{12}(\infty)}{\sqrt{2\pi}}A_2(r)+\frac{1}{\sqrt{2\pi}}\int_r^\infty u^{-3/2}e^{-c_2^2u/2}\{J_{12}(u)-J_{12}(\infty)\}\,du.$$

The second term is $o(A_2(r))$ because

$$\sup_{u\ge r}|J_{12}(u)-J_{12}(\infty)|\xrightarrow{r\to\infty}0$$

and hence

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big)=\frac{J_{12}(\infty)+o(1)}{\sqrt{2\pi}}A_2(r).$$

Finally we use

$$A_2(r)=\int_r^\infty u^{-3/2}e^{-c_2^2u/2}\,du\sim\frac{2}{c_2^2}r^{-3/2}e^{-c_2^2r/2}$$

to conclude that

$$\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0),Q_{W_2}(r)\big)\sim K_{12}r^{-3/2}e^{-c_2^2r/2}$$

with

$$K_{12}:=\frac{\sqrt{2/\pi}}{c_2^2}\int_0^\infty ae^{c_2a}L_{12}(a)\,da.$$

It remains to calculate the constant K_{12}. We have

$$\begin{aligned}
J_{12}(\infty)&=\int_0^\infty ae^{c_2a}\mathbb{E}\Big\{(Q_{W_1}(0)-m_1)\mathbf{1}_{\{Q_{W_2}(0)>a\}}\Big\}\,da\\
&=\mathbb{E}\left\{(Q_{W_1}(0)-m_1)\int_0^{Q_{W_2}(0)}ae^{c_2a}\,da\right\}\\
&=\frac{1}{c_2^2}\mathbb{E}\Big\{(Q_{W_1}(0)-m_1)\Big\{e^{c_2Q_{W_2}(0)}(c_2Q_{W_2}(0)-1)+1\Big\}\Big\}\\
&=\frac{1}{c_2^2}\mathbb{E}\Big\{(Q_{W_1}(0)-m_1)e^{c_2Q_{W_2}(0)}(c_2Q_{W_2}(0)-1)\Big\},
\end{aligned}$$

where in the last equality we used that $\mathbb{E}\{Q_{W_1}(0)-m_1\}=0$.
Hence

$$K_{12}=\frac{\sqrt{2/\pi}}{c_2^4}\mathbb{E}\Big\{(Q_{W_1}(0)-m_1)e^{c_2Q_{W_2}(0)}(c_2Q_{W_2}(0)-1)\Big\}.$$

Define

$$F_2(x):=e^{c_2x}(c_2x-1),\qquad x\ge 0.$$

Since $Q_{W_2}(0) \sim \mathrm{Exp}(2c_2)$,

$$\mathbb{E}\{F_2(Q_{W_2}(0))\} = \int_0^\infty e^{c_2 x}(c_2 x - 1)2c_2 e^{-2c_2 x}\,dx = 0.$$

Thus, it follows that

$$\mathbb{E}\Big\{(Q_{W_1}(0) - m_1)e^{c_2 Q_{W_2}(0)}(c_2 Q_{W_2}(0) - 1)\Big\} = \mathbb{E}\{(Q_{W_1}(0) - m_1)F_2(Q_{W_2}(0))\} = \mathbb{E}\{Q_{W_1}(0)F_2(Q_{W_2}(0))\}.$$

For $c > 0$, let

$$M_c := \sup_{t\ge 0}(B(t) - ct), \qquad S_x^{(c)} := \inf\{t \ge 0 : B(t) - ct = x\}, \quad x > 0.$$

The pair $(Q_{W_1}(0), Q_{W_2}(0))$ has the same law as (M_{c_1}, M_{c_2}), so we may use the latter pair to calculate the expectations above. Following, e.g., [24, Section 3.5.C, p. 197, Eq. (5.12)], we have

$$\mathbb{E}\Big\{e^{qS_x^{(c)}};\ S_x^{(c)} < \infty\Big\} = e^{-x(c+\sqrt{c^2-2q})}, \qquad x > 0, \quad q < c^2/2. \tag{36}$$

Thus, differentiating with respect to q (which is justified by dominated convergence, since the transform is finite in a neighborhood of every $q < c^2/2$), we obtain

$$\mathbb{E}\Big\{S_x^{(c)}e^{qS_x^{(c)}};\ S_x^{(c)} < \infty\Big\} = \frac{x}{\sqrt{c^2-2q}}\,e^{-x(c+\sqrt{c^2-2q})}, \qquad q < c^2/2. \tag{37}$$

Suppose first that $0 < c_1 \le c_2$. Under $\{S_x^{(c_2)} < \infty\}$, let E_{c_1} be the maximum of the Brownian increments after $S_x^{(c_2)}$, with drift $-c_1$. By the strong Markov property, $E_{c_1} \sim \mathrm{Exp}(2c_1)$ and is independent of the pre-hitting path. Moreover,

$$M_{c_1} = x + (c_2 - c_1)S_x^{(c_2)} + E_{c_1}.$$

Indeed, before $S_x^{(c_2)}$, $B(t) - c_1 t \le x + (c_2 - c_1)t \le x + (c_2 - c_1)S_x^{(c_2)}$, so the global maximum is attained at or after this hitting time. Also, $\{M_{c_2} > x\} = \{S_x^{(c_2)} < \infty\}$ up to a null set for each $x > 0$.
Since $\mathbb{E}\{E_{c_1}\} = m_1$, applying (36) and (37) with $c = c_2$ and $q = 0$ gives

$$\begin{aligned} L_{12}(x) &= \mathbb{E}\Big\{(M_{c_1} - m_1)\mathbf{1}_{\{M_{c_2}>x\}}\Big\} \\ &= x\mathbb{P}\{S_x^{(c_2)} < \infty\} + (c_2 - c_1)\mathbb{E}\Big\{S_x^{(c_2)};\ S_x^{(c_2)} < \infty\Big\} \\ &= \frac{2c_2 - c_1}{c_2}\,xe^{-2c_2 x}. \end{aligned}$$

Consequently,

$$K_{12} = \frac{\sqrt{2/\pi}}{c_2^2}\frac{2c_2 - c_1}{c_2}\int_0^\infty x^2 e^{-c_2 x}\,dx = \sqrt{\frac{2}{\pi}}\frac{2(2c_2 - c_1)}{c_2^6}.$$

If $c_1 > c_2$, let

$$d := c_1 - c_2, \qquad k := \sqrt{d^2 + c_2^2}.$$

An elementary integration against the $\mathrm{Exp}(2c_2)$ density yields

$$\mathbb{E}\{F_2(y + E_{c_2})\} = 2c_2 y e^{c_2 y}, \qquad y \ge 0.$$

The same Hölder estimate as used in (35) yields $\mathbb{E}\{M_{c_1}|F_2(M_{c_2})|\} < \infty$, and we have

$$\begin{aligned} \mathbb{E}\{Q_{W_1}(0)F_2(Q_{W_2}(0))\} &= \int_0^\infty \mathbb{E}\Big\{F_2(M_{c_2})\mathbf{1}_{\{M_{c_1}>x\}}\Big\}\,dx \\ &= 2c_2\int_0^\infty e^{c_2 x}\mathbb{E}\Big\{(x + dS_x^{(c_1)})e^{c_2 dS_x^{(c_1)}};\ S_x^{(c_1)} < \infty\Big\}\,dx \\ &= 2c_2\left(1 + \frac{d}{k}\right)\int_0^\infty xe^{-(d+k)x}\,dx \\ &= \frac{2c_2}{k(d+k)} = \frac{2}{c_2}\left(1 - \frac{d}{k}\right). \end{aligned}$$

Finally,

$$K_{12}=\sqrt{\frac{2}{\pi}}\begin{cases}\dfrac{2(2c_2-c_1)}{c_2^6}, & 0<c_1\le c_2\\ \dfrac{2}{c_2^5}\left(1-\dfrac{c_1-c_2}{\sqrt{(c_1-c_2)^2+c_2^2}}\right), & c_1>c_2\end{cases}.$$

This completes the proof. □

4.2. **Proof of Theorem 2.3.** Let $\varepsilon\in\{-1,1\}$ be the eventual sign of V_X''. For the proof, we take the positive normalisation $b(r)=|V_X''(r)|$ for all sufficiently large r, and extend it arbitrarily to a positive function on bounded intervals. Then b is regularly varying with index $-\alpha$, and $V_X''(r)=\varepsilon b(r)$ for all large r.
Suppose that $\alpha>0$, and let $\delta\in(0,\min\{\alpha,1\})$. Potter theorem [5, Section 1.5] implies that $b(s)\le Cs^{-\delta}$ for all $s\ge r_0$, where $r_0>t_0$ is sufficiently large. Twice integrating V_X'' then yields

$$0\le V_X(t)\le C(1+t)+C\int_{r_0}^{t}(t-s)s^{-\delta}\,ds=O(t^{2-\delta}),\qquad t\to\infty.$$

Thus (30) holds, and $b(r)\xrightarrow{r\to\infty}0$.
If $\alpha=0$, twice integrating V_X'' from a sufficiently large $r_0>t_0$ and applying Karamata's theorem [5, Section 1.5] yields

$$V_X(r)\sim\frac{\varepsilon}{2}r^2b(r),\qquad r\to\infty.$$

Since $V_X\ge0$, necessarily $\varepsilon=1$. The assumption $V_X''(r)\log r\xrightarrow{r\to\infty}0$ implies (30) and $b(r)\xrightarrow{r\to\infty}0$.
The above implies that all the claims in Lemma 3.11 are satisfied.
We shall apply Theorem 3.8 to the workload fields in (33). Their cross-covariance is

$$\begin{aligned}R_r(u,v)&:=\mathbb{C}\mathrm{ov}(\mathcal{X}(u),\mathcal{Y}_r(v))\\&=\frac{1}{2}\{V_X(r+u)+V_X(|r-v|)-V_X(|r+u-v|)-V_X(r)\}.\end{aligned}$$

For all sufficiently large r and $v\le r/2$, the entire range $r+s-t$, $0\le s\le u$, $0\le t\le v$, lies in the interval of twice continuous differentiability of V_X. Hence

$$R_r(u,v)=\frac{1}{2}\int_0^u\int_0^v V_X''(r+s-t)\,dt\,ds.\tag{38}$$

For every fixed $T>0$, the uniform convergence theorem for regular variation implies

$$\sup_{|x|\le T}\left|\frac{V_X''(r+x)}{b(r)}-\varepsilon\right|\xrightarrow{r\to\infty}0.$$

Thus $K_r(u,v):=R_r(u,v)/b(r)\xrightarrow{r\to\infty}\varepsilon uv/2$ uniformly on every compact square, so the limiting functions in condition (27) of Theorem 3.8 are $g_1(u)=u$ and $g_2(v)=\varepsilon v/2$.
Next, we identify functions $G_{\mathcal{X}},G_{\mathcal{Y}}$ that satisfy condition (20) in Lemma 3.7. Put

$$G_{\mathcal{X}}(x)=G_{\mathcal{Y}}(x)=G(x):=\left(\sup_{0\le t\le x}V_X(t)\right)^{1/2}.$$

This function is nonnegative and nondecreasing, and Cauchy–Schwarz yields

$$|R_r(u,v)|\le\sqrt{V_X(u)V_X(v)}\le G(u)G(v),\qquad r,u,v\ge0,$$

which confirms that (21) in Lemma 3.7 is satisfied.
Let $\tau_{\mathcal{X}^{(\theta)}}$ and $\tau_{\mathcal{Y}_r^{(\theta)}}$ be the maximising times of the interpolated fields $\mathcal{X}^{(\theta)},\mathcal{Y}_r^{(\theta)}$. Their marginal laws are those of τ_{W_1},τ_{W_2}. Thus,

$$\sup_{r,\theta}\mathbb{E}\{G(\tau_{\mathcal{X}^{(\theta)}})G(\tau_{\mathcal{Y}_r^{(\theta)}})\}\le\left(\mathbb{E}\{G(\tau_{W_1})^2\}\right)^{1/2}\left(\mathbb{E}\{G(\tau_{W_2})^2\}\right)^{1/2}<\infty.$$

We now verify normalised localisation (23). For $v \le r/2$, put $x = r + s - t$ in (38). Then $x/r \ge 1/2$ and $x/r \le 1 + u/r$. Potter bounds [5, Section 1.5] yield, for all large r,

$$\frac{|V_X''(x)|}{b(r)} \le C \max\{(x/r)^{-\alpha+1}, (x/r)^{-\alpha-1}\} \le C(1+u).$$

Consequently,

$$|K_r(u,v)| \le Cu(1+u)v, \qquad v \le r/2.$$

For $v > r/2$, using that $G(x) \le C(1+x)$ we get

$$|R_r(u,v)| \le G(u)G(v) \le C(1+u)(1+v).$$

Since $1/b(r)$ is regularly varying with index α, it is bounded by $Cr^{\alpha+1}$ for all large r. Therefore

$$|K_r(u,v)| \le C(1+u)(1+v)r^{\alpha+1} \le C(1+u)(1+v)^{\alpha+2}.$$

Combining the two regions yields

$$|K_r(u,v)| \le C(1+u)^2(1+v)^{\alpha+2}, \qquad u,v \ge 0,$$

uniformly for all large r.

For $T > 0$ and $\theta \in [0,1]$, define the normalised truncation error as in (23) in Lemma 3.7:

$$\begin{aligned} J_{r,\theta}(T) := &\left| K_r\big(\tau_{\mathcal{X}^{(\theta)}}, \tau_{\mathcal{Y}_r^{(\theta)}}\big) - K_r\big(\tau_{\mathcal{X}^{(\theta)}}(T), \tau_{\mathcal{Y}_r^{(\theta)}}\big)\right| \\ &+ \left| K_r\big(\tau_{\mathcal{X}^{(\theta)}}(T), \tau_{\mathcal{Y}_r^{(\theta)}}\big) - K_r\big(\tau_{\mathcal{X}^{(\theta)}}(T), \tau_{\mathcal{Y}_r^{(\theta)}}(T)\big)\right|, \end{aligned}$$

where $\tau_{\mathcal{X}^{(\theta)}}(T)$ and $\tau_{\mathcal{Y}_r^{(\theta)}}(T)$ denote the maximising times over $[0,T]$. Since $K_r = R_r/b(r)$, condition (23) is equivalent to

$$\lim_{T\to\infty} \limsup_{r\to\infty} \sup_{\theta\in[0,1]} \mathbb{E}[J_{r,\theta}(T)] = 0.$$

For each interpolated field, the maximising time over $[0,T]$ is no greater than its global maximising time, and the two coincide whenever the global maximising time belongs to $[0,T]$. Consequently, the first difference defining $J_{r,\theta}(T)$ vanishes on $\tau_{\mathcal{X}^{(\theta)}} \le T$, and the second vanishes on $\tau_{\mathcal{Y}_r^{(\theta)}} \le T$.

By (25),

$$J_{r,\theta}(T) \le C(1+\tau_{\mathcal{X}^{(\theta)}})^2(1+\tau_{\mathcal{Y}_r^{(\theta)}})^{\alpha+2}\big(\mathbf{1}_{\{\tau_{\mathcal{X}^{(\theta)}}>T\}} + \mathbf{1}_{\{\tau_{\mathcal{Y}_r^{(\theta)}}>T\}}\big).$$

Applying Cauchy–Schwarz separately to the two terms in the preceding bound, and using $\tau_{\mathcal{X}^{(\theta)}} \overset{d}{=} \tau_{W_1}$, $\tau_{\mathcal{Y}_r^{(\theta)}} \overset{d}{=} \tau_{W_2}$, we obtain, for all sufficiently large r and every $\theta \in [0,1]$,

$$\begin{aligned} \mathbb{E}\{J_{r,\theta}(T)\} \le\ & C\left(\mathbb{E}\Big\{(1+\tau_{W_1})^4\mathbf{1}_{\{\tau_{W_1}>T\}}\Big\}\right)^{1/2}\left(\mathbb{E}\{(1+\tau_{W_2})^{2\alpha+4}\}\right)^{1/2} \\ &+ C\left(\mathbb{E}\{(1+\tau_{W_1})^4\}\right)^{1/2}\left(\mathbb{E}\Big\{(1+\tau_{W_2})^{2\alpha+4}\mathbf{1}_{\{\tau_{W_2}>T\}}\Big\}\right)^{1/2}. \end{aligned} \tag{39}$$

By Lemma 3.11, τ_{W_i} has finite moments of every positive order. In particular,

$$\mathbb{E}\{(1+\tau_{W_1})^4\} < \infty, \qquad \mathbb{E}\{(1+\tau_{W_2})^{2\alpha+4}\} < \infty.$$

Hence dominated convergence implies that the right-hand side of (39) tends to zero as $T \to \infty$ uniformly for sufficiently large r and $\theta \in [0,1]$, implying that (23) holds.

Finally, $0 \le \tau_{W_i}(T) \le \tau_{W_i}$, and compact and global maximisers agree once $T \ge \tau_{W_i}$. Their finite first moments therefore imply the requested L^1 convergence (28) in Theorem 3.8 (recall that $g_1(u) = u$ and $g_2(v) = \varepsilon v/2$).

Thus, all conditions of Theorem 3.8 are verified, which leads to

$$\lim_{r\to\infty} \frac{\mathbb{C}\mathrm{ov}\big(Q_{W_1}(0), Q_{W_2}(r)\big)}{b(r)} = \frac{\varepsilon}{2}\mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\}.$$

For any $u > 0$, $V_X(u) > 0$ implies $\mathbb{P}\{X(u) - c_i u > 0\} > 0$, and hence $\mathbb{P}\{\tau_{W_i} > 0\} > 0$. Therefore $\varepsilon\mathbb{E}\{\tau_{W_1}\}\mathbb{E}\{\tau_{W_2}\}/2 \neq 0$. Since $\varepsilon b(r) = V_X''(r)$ for all large r, this proves the stated asymptotic equivalence.

□

4.3. **Proof of Theorem 2.6.** First, we establish the variance growth and maximising-time estimates. Choose $t_1 > t_0$ sufficiently large that

$$|V_X''(t)| \le Ce^{-\lambda t}, \qquad t \ge t_1.$$

The integrability of V_X'' at infinity implies that $V_X'(t) \xrightarrow{t\to\infty} D_X$ for some finite constant D_X. Moreover,

$$V_X'(t) - D_X = -\int_t^\infty V_X''(s)\,ds = O(e^{-\lambda t}),$$

so $V_X(t) - D_X t$ converges to a finite constant B_X. Integrating the last identity yields

$$V_X(t) = D_X t + B_X + \int_t^\infty (s-t)V_X''(s)\,ds = D_X t + B_X + \frac{a}{\lambda^2}e^{-\lambda t}(1+o(1)). \tag{40}$$

Since $V_X(t) \ge 0$, then $D_X \ge 0$, and $V_X(t)/t \xrightarrow{t\to\infty} D_X$. In particular, $V_X(t) \le C(1+t)$ and (30) holds. By Lemma 3.11, the processes W_i tend to $-\infty$, have almost surely unique compact and global maximisers, and M_{W_i}, τ_{W_i} have all positive moments. Furthermore,

$$\mathbb{E}\{e^{q\tau_{W_i}}\} < \infty, \qquad 0 < q < \frac{c_i^2}{2D_X}, \qquad i = 1, 2,$$

with the upper endpoint interpreted as $+\infty$ if $D_X = 0$. Condition (11) therefore yields $\mathbb{E}\{e^{\lambda\tau_{W_2}}\} < \infty$. Similarly to the proof of Theorem 2.6, for the fields in (33), we introduce

$$\begin{aligned} R_r(u,v) &:= \mathbb{C}\mathrm{ov}(\mathcal{X}(u), \mathcal{Y}_r(v)) \\ &= \frac{1}{2}\{V_X(r+u) + V_X(|r-v|) - V_X(|r+u-v|) - V_X(r)\} \end{aligned} \tag{41}$$

and set $b(r) = e^{-\lambda r}$, and $K_r = R_r/b(r)$. For $r \ge t_1$, $u \ge 0$, and $0 \le v \le r - t_1$,

$$R_r(u,v) = \frac{1}{2}\int_0^u \int_0^v V_X''(r+s-t)\,dt\,ds. \tag{42}$$

For every fixed $T > 0$, assumption (10) implies

$$\sup_{|x|\le T} \left| e^{\lambda r} V_X''(r+x) - ae^{-\lambda x} \right| \xrightarrow{r\to\infty} 0.$$

Thus, uniformly on $[0,T]^2$,

$$K_r(u,v) \xrightarrow{r\to\infty} \frac{a}{2\lambda^2}(1-e^{-\lambda u})(e^{\lambda v} - 1).$$

Hence, condition (27) of Theorem 3.8 holds with

$$g_1(u) = 1 - e^{-\lambda u}, \qquad g_2(v) = \frac{a}{2\lambda^2}(e^{\lambda v} - 1).$$

For checking (20), we put

$$G(x) := \left(\sup_{0\le t\le x} V_X(t) \right)^{1/2},$$

and observe that G is nonnegative and nondecreasing, $G(x) \le C(1+x)^{1/2}$. Thus,

$$|R_r(u,v)| \le \sqrt{V_X(u)V_X(v)} \le G(u)G(v). \tag{43}$$

Recalling that $\tau_{\mathcal{X}^{(\theta)}} \stackrel{d}{=} \tau_{W_1}$ and $\tau_{\mathcal{Y}_r^{(\theta)}} \stackrel{d}{=} \tau_{W_2}$, Cauchy–Schwarz, and (43) imply

$$\begin{aligned} &\sup_{r\ge 0,\ \theta\in[0,1]} \mathbb{E}\Big\{ G(\tau_{\mathcal{X}^{(\theta)}}) G(\tau_{\mathcal{Y}_r^{(\theta)}}) \Big\} \\ &\quad \le \left(\mathbb{E}\{G(\tau_{W_1})^2\}\right)^{1/2} \left(\mathbb{E}\{G(\tau_{W_2})^2\}\right)^{1/2} \\ &\quad \le C\left(1 + \mathbb{E}\{\tau_{W_1}\}\right)^{1/2} \left(1 + \mathbb{E}\{\tau_{W_2}\}\right)^{1/2} < \infty. \end{aligned} \tag{44}$$

The last inequality follows from Lemma 3.11. Together, (43) and (44) verify the domination and product integrability assumptions of Lemma 3.7.

In order to verify (23), we first prove

$$|K_r(u,v)| \le Ce^{\lambda v}, \qquad r \ge t_1, \quad u,v \ge 0, \tag{45}$$

with C independent of r, u, v.
If $v \le r - t_1$, then (42), $K_r = e^{\lambda r} R_r$, and $|V_X''(x)| \le Ce^{-\lambda x}$ for $x \ge t_1$ give

$$\begin{aligned} |K_r(u,v)| &\le C \int_0^u \int_0^v e^{-\lambda(s-t)}\, dt\, ds \\ &= \frac{C}{\lambda^2}(1 - e^{-\lambda u})(e^{\lambda v} - 1) \le Ce^{\lambda v}. \end{aligned}$$

If $v > r - t_1$, set $h(t) := V_X(t) - D_X t$, $t \ge 0$. By continuity and (40),

$$H := \sup_{t \ge 0} |h(t)| < \infty.$$

Substituting $V_X(t) = D_X t + h(t)$ into (41), and using

$$\frac{D_X}{2}\{u + |r - v| - |r + u - v|\} = D_X \min\{u, (v - r)_+\},$$

we obtain

$$|R_r(u,v) - D_X \min\{u, (v-r)_+\}| \le 2H.$$

Since $D_X \ge 0$, it follows that

$$|R_r(u,v)| \le D_X(v - r)_+ + 2H.$$

Writing $d = v - r > -t_1$, we therefore have

$$\begin{aligned} |K_r(u,v)| &\le e^{\lambda r}\{D_X(v-r)_+ + 2H\} \\ &= e^{\lambda v} e^{-\lambda d}\{D_X d_+ + 2H\} \le Ce^{\lambda v}, \end{aligned}$$

where

$$C := \sup_{d > -t_1} e^{-\lambda d}\{D_X d_+ + 2H\} < \infty.$$

Thus, (45) holds.
Next, in order to prove (23) in Lemma 3.7, we analyze

$$J_{r,\theta}(T) := \left|K_r\big(\tau_{\mathcal{X}^{(\theta)}}, \tau_{\mathcal{Y}_r^{(\theta)}}\big) - K_r\big(\tau_{\mathcal{X}^{(\theta)}}(T), \tau_{\mathcal{Y}_r^{(\theta)}}\big)\right| + \left|K_r\big(\tau_{\mathcal{X}^{(\theta)}}(T), \tau_{\mathcal{Y}_r^{(\theta)}}\big) - K_r\big(\tau_{\mathcal{X}^{(\theta)}}(T), \tau_{\mathcal{Y}_r^{(\theta)}}(T)\big)\right| \tag{46}$$

for $T > 0$ and $\theta \in [0,1]$. Since $K_r = R_r/b(r)$, it suffices to prove

$$\lim_{T \to \infty} \sup_{r \ge t_1,\, \theta \in [0,1]} \mathbb{E}\{J_{r,\theta}(T)\} = 0.$$

Clearly, the first difference in (46) vanishes on $\{\tau_{\mathcal{X}^{(\theta)}} \le T\}$, and the second vanishes on $\{\tau_{\mathcal{Y}_r^{(\theta)}} \le T\}$. Moreover, by (45) and the monotonicity of $v \mapsto e^{\lambda v}$, we obtain

$$J_{r,\theta}(T) \le Ce^{\lambda \tau_{\mathcal{Y}_r^{(\theta)}}} \left(\mathbf{1}_{\{\tau_{\mathcal{X}^{(\theta)}} > T\}} + \mathbf{1}_{\{\tau_{\mathcal{Y}_r^{(\theta)}} > T\}}\right), \qquad r \ge t_1. \tag{47}$$

Recalling that $\tau_{\mathcal{X}^{(\theta)}} \overset{d}{=} \tau_{W_1}$ and $\tau_{\mathcal{Y}_r^{(\theta)}} \overset{d}{=} \tau_{W_2}$., by (31) and (11), we know that $\mathbb{E}\{e^{\lambda \tau_{W_2}}\} < \infty$. Hence,

$$\sup_{r \ge t_1,\, \theta \in [0,1]} \mathbb{E}\left\{e^{\lambda \tau_{\mathcal{Y}_r^{(\theta)}}} \mathbf{1}_{\{\tau_{\mathcal{Y}_r^{(\theta)}} > T\}}\right\} = \mathbb{E}\left\{e^{\lambda \tau_{W_2}} \mathbf{1}_{\{\tau_{W_2} > T\}}\right\} \xrightarrow{T \to \infty} 0.$$

For the analysis of the other term in (47), let us fix $L > 0$. Splitting according to whether $e^{\lambda \tau_{\mathcal{Y}_r^{(\theta)}}} > L$ leads to

$$\sup_{r \ge t_1,\, \theta \in [0,1]} \mathbb{E}\left\{e^{\lambda \tau_{\mathcal{Y}_r^{(\theta)}}} \mathbf{1}_{\{\tau_{\mathcal{X}^{(\theta)}} > T\}}\right\} \le \mathbb{E}\left\{e^{\lambda \tau_{W_2}} \mathbf{1}_{\{e^{\lambda \tau_{W_2}} > L\}}\right\} + L\mathbb{P}\{\tau_{W_1} > T\}.$$

Since $\tau_{W_1} < \infty$ almost surely by Lemma 3.11, letting $T \to \infty$ with L fixed yields

$$\limsup_{T \to \infty} \sup_{r \ge t_1,\, \theta \in [0,1]} \mathbb{E}\left\{e^{\lambda \tau_{\mathcal{Y}_r^{(\theta)}}} \mathbf{1}_{\{\tau_{\mathcal{X}^{(\theta)}} > T\}}\right\} \le \mathbb{E}\left\{e^{\lambda \tau_{W_2}} \mathbf{1}_{\{e^{\lambda \tau_{W_2}} > L\}}\right\}.$$

By the dominated convergence theorem, the right-hand side of the above tends to zero as $L \to \infty$. Thus both terms in the bound (47) tend to zero uniformly in $r \geq t_1$ and $\theta \in [0,1]$, proving (23). Finally, $0 \leq \tau_{W_i}(T) \leq \tau_{W_i}$, and these locations agree once $T \geq \tau_{W_i}$. Boundedness of g_1 and integrability of $g_2(\tau_{W_2})$ yield the L^1 convergence in (28).

To sum up, all the conditions of Theorem 3.8 are satisfied, yielding the limit in (12) after division by $e^{-\lambda r}$.

We also note that, for any $u > 0$, the assumption $V_X(u) > 0$ implies $\mathbb{P}\{X(u) - c_i u > 0\} > 0$, and hence $\mathbb{P}\{\tau_{W_i} > 0\} > 0$. Thus, $\mathbb{E}\{1 - e^{-\lambda \tau_{W_1}}\}, \mathbb{E}\{e^{\lambda \tau_{W_2}} - 1\} > \in (0, \infty)$. This completes the proof. □

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